

GrainTrust AI: The AI-Powered Escrow & Supply Chain Network for Rice Milling

What problem will your idea/ project solve?

The Nigerian rice value chain is broken by a massive "Trust and Liquidity Deficit" between rural smallholder farmers and large industrial milling companies.

1. **Delayed Payments:** Large mills typically take 14-30 days to process payments for delivered rice paddies. Rural farmers cannot wait, so they sell their crops to exploitative middlemen at a 30% discount just to get instant cash.
2. **Quality Disputes:** Mills often reject deliveries or underpay farmers, claiming the rice has high moisture or impurities, leading to disputes.
3. **Supply Chain Blindness:** Rice mills lack predictive data on national farm yields, causing factory downtime when raw materials run short.

GrainTrust AI solves this by bridging Computer Vision AI with Interswitch's Payment Infrastructure to create a "Trustless, Instant-Settlement Supply Chain."

Describe your project idea in 100 words or less

GrainTrust AI is a B2B agricultural platform that guarantees predictable supply for rice mills and instant payments for farmers. Using our mobile app, cooperative agents capture field data; our AI predicts the rice yield and alerts milling companies. The mill issues a "Digital Purchase Order" by funding an **Interswitch Escrow Account**. When the rice arrives at the factory, our AI Quality-Control camera scans the paddy for impurities and moisture. If the AI approves the quality, it automatically triggers the **Interswitch Disbursement API**, instantly crediting the farmer's bank account. We eliminate middlemen, guarantee crop quality, and digitize agricultural payments.

The 4 Pillars of the GrainTrust AI Ecosystem

Pillar 1: AI Yield Prediction (The Field App)

- **The Action:** Local cooperative agents use the GrainTrust mobile app to register rice farms and snap photos of the fields nearing harvest.
- **The AI Assessment:** The AI estimates the harvest date and the expected tonnage (e.g., 10 Tons of Faro 44 Rice).
- **The Dashboard Sync:** This predictive data is beamed to the Central Command Dashboard of participating Rice Milling Companies, allowing them to see exactly how much rice is ready for purchase nationwide.

Pillar 2: The Interswitch Escrow (Trust Generation)

- **The Contract:** A milling company sees 10 tons of rice ready in Kano. They click "Secure Harvest" on their dashboard.

- **Interswitch Integration:** The system redirects the mill to the **Interswitch Payment Gateway (IPG)** to deposit the funds (e.g., ₦5,000,000) into a digital Escrow account.
- **The SMS Alert:** The farmer instantly receives an SMS: *"Your 10 Tons of rice has been booked by MegaMills. ₦5,000,000 is securely locked in Interswitch Escrow. Please dispatch the truck."* * **Impact:** The farmer now has 100% confidence to bypass the middleman and send the truck directly to the factory, knowing the money is guaranteed.

Pillar 3: AI Quality Control (The Factory Gate)

- **The Problem:** The mill only wants to pay for high-quality, dry rice paddies without stones or mold.
- **The Action:** When the truck arrives at the factory, a sample is placed under the GrainTrust AI Camera.
- **The AI Grading:** The Computer Vision model instantly analyzes the grain size, discoloration, and impurity ratio, grading it (e.g., Grade A - 98% Purity).

Pillar 4: Automated Interswitch Disbursement (The Settlement)

- **Instant Payout:** The moment the AI validates that the rice meets the Grade A standard, the GrainTrust backend fires an API call to **Interswitch's Disbursement/Transfer API**.
- **The Magic:** The ₦5,000,000 is instantly released from the Escrow and credited directly into the rural farmer's local bank account or mobile wallet.
- **Impact:** A transaction that used to take 30 days of paperwork and corporate approvals now happens in 3 seconds at the factory gate.

Why GrainTrust AI Will Win the Interswitch Buildathon

1. **Perfect Theme Alignment (Agriculture + Payments):** It completely disrupts how agricultural commodities are traded by using Interswitch to solve the biggest pain point in farming: delayed payments.
2. **Innovative Use of APIs:** It utilizes multiple Interswitch layers—the Payment Gateway (IPG) for the corporate escrow deposit, and the Disbursement API for the instant farmer payout.
3. **High Technical Feasibility:** The AI vision models for grain analysis are readily available, and integrating Interswitch APIs into a Streamlit/Python backend can easily be prototyped within the 7-day hackathon window.
4. **Massive Market Impact:** By eliminating middlemen who exploit payment delays, farmers increase their profit margins by 30%, while corporate mills secure a steady, high-quality supply chain.

Upload a 2 mins video (Video Pitch Script)

[0:00 - 0:30] The Problem: "Hello Judges. In Nigeria, rural rice farmers lose up to 30% of their profits to middlemen. Why? Because large milling companies take 30 days to process payments, and farmers need cash instantly to survive. This breaks the supply chain. Milling factories run out of rice, and farmers stay poor. Our solution is GrainTrust AI."

[0:30 - 1:00] The AI & Escrow Solution: "GrainTrust AI bridges Computer Vision with Interswitch's payment infrastructure to create an instant-settlement supply chain. First, our AI

field app predicts a farmer's harvest. A milling company sees this on our dashboard and secures the crop by funding a digital Escrow via the Interswitch Payment Gateway. The farmer gets an SMS that the money is secured, giving them the trust to ship directly to the factory."

[1:00 - 1:30] The Automated Settlement: "When the truck arrives, our AI Camera scans the rice to guarantee quality—checking for impurities and moisture. The moment the AI passes the quality check, our system triggers the Interswitch Disbursement API. The funds are instantly released from escrow into the farmer's bank account before the truck even turns off its engine."

[1:30 - 2:00] The Value Proposition: "No more 30-day delays. No more quality disputes. No more middlemen. Just predictable supply for mills, secured by AI, and instant wealth for farmers, powered by Interswitch. Thank you."